

Transradial artery access for carotid artery stenting: A pooled analysis

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Abstract

Introduction: Carotid artery stenting (CAS) through transradial access (TRA) is emerging as an alternative to carotid endarterectomy. However, the current evidence base is limited, mainly comprising single-center studies.

Objective: This systematic review and meta-analysis aim to assess the safety and effectiveness of TRA for CAS, providing evidence to support clinical decisions.

Methods: We conducted searches on PUBMED, Cochrane Library, Embase, and Web of Science databases, including studies on TRA for CAS. Studies with fewer than 20 patients, non-primary outcomes, and non-full-text articles were excluded.

Results: We analyzed 14 studies involving 1,166 patients who underwent CAS via TRA. Procedural success rate was high in 13 studies, with a 95% rate (95% CI: 92%–98%). Crossover to TFA access was observed in 12 studies at 6% (95% CI: 3%–9%). Transradial access failure was reported in four studies, with a rate of 0% (95% CI: 0%–0%). Cannulation failure resulted in a rate of 4% (95% CI: 2%–7%). Asymptomatic radial artery occlusion (ARAO) occurred at a rate of 2% based on eight studies (95% CI: 0%–5%). Forearm hematoma was reported in 10 studies, with an occurrence of 1% (95% CI: 0%–2%). Cerebral vascular attacks (CAV) within 30 days were assessed in 13 studies, indicating a 2% occurrence (95% CI: 1%–2%).

Conclusion: The findings suggest that TRA for CAS yields promising outcomes with high success rates and low complication rates. Further research should focus on randomized controlled trials and long-term outcomes to validate and extend findings.

Keywords

Radial, transradial, transradial access, carotid, carotid artery stenting, carotid artery stenting

Introduction

Carotid artery stenting (CAS) has emerged as a less invasive alternative to carotid endarterectomy for the treatment of carotid artery stenosis.¹ While transfemoral access (TFA) has traditionally been the preferred approach for CAS procedures, radial artery access has gained increasing attention as a potentially safer and more patient-friendly alternative.²

Transradial access (TRA) involves accessing the carotid artery through the radial artery in the wrist instead of using the femoral artery in the groin.³ This technique offers several potential benefits. Firstly, it provides a superficial and easily accessible site for arterial access, reducing the risk of complications associated with deep groin access, such as bleeding, hematoma, or infection. Secondly, the radial artery, due to its smaller diameter compared to the femoral artery, may potentially reduce the risk of bleeding events during the procedure. Additionally, the radial artery approach offers enhanced patient comfort and satisfaction, as it avoids the need for groin manipulation and allows for earlier mobilization after the procedure.^{4–8}

Despite these potential advantages, the radial artery approach for CAS is challenging. One of the main challenges is the learning curve associated with mastering the technique, as it requires specific skills and training to successfully navigate the tortuous anatomy from the wrist to the carotid artery.⁹

Although the results seem to be promising, it is important to remember that complications are present, even at a low rate.^{10,11}

Furthermore, the radial artery can be subject to spasms, making it necessary to use vasodilators to facilitate access, as well as anatomical variations or limitations in specific patient populations that could affect the feasibility and success of the radial artery approach.^{12,13} Additionally, the adoption of TRA for CAS remains a subject of debate. A limited body of evidence focuses explicitly on TRA for CAS, consisting mainly of single-center studies.^{14–16} Therefore, conducting a

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pooled analysis that systematically evaluates the available data can provide a more comprehensive assessment of the safety and efficacy outcomes of this novel access approach. By synthesizing the results from these studies, we aim to generate more robust evidence to inform clinical decision-making about this topic.

Methods

Eligibility criteria

This systematic review included all studies that reported TRA for CAS. To minimize the risk of bias, studies with fewer than 20 patients, where the primary outcome was not found, and letters, comments, and abstracts were excluded.

Search strategy and data extraction

We systematically searched on the PUBMED, Cochrane Library, Embase, and Web of Science databases. Our search strategy was carefully crafted to conduct a thorough investigation of the topic, utilizing a comprehensive combination of relevant keywords. The specific keywords employed in our search included: “carotid,” “radial,” “transradial,” “TRA,” “stent,” and “stenting.” This meticulous approach ensured that we obtained the most pertinent and reliable information, empowering us to present a well-founded and in-depth analysis of the subject matter. Two authors (J.B and S.B) independently extracted the data following predefined search criteria.

Risk of bias assessment

The quality of the case-control studies eligible for inclusion was evaluated using the Newcastle-Ottawa Scale (NOS). Studies with the NOS scores 0–3, 4–6, and 7–9 were considered low, moderate, and high quality, respectively. Low-quality studies were excluded due to the high risk of bias.

Statistical analysis

This systematic review was performed per the Cochrane Collaboration and the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) statement guidelines. Relative risk (RR) with 95% confidence intervals was used to compare outcome treatment effects. I^2 statistics were used to assess for heterogeneity; p -value inferior to 0.05 and $I^2 < 35\%$ were considered significant. Statistical analysis was performed using the software R (version 4.2.3, R Foundation for Statistical Computing, Vienna, Austria).

Results

Study selection

We found 1,775 articles, 406 in PubMed, 922 in Embase, 420 in Web of Science, and 27 in Cochrane databases. A total of 1,016 non-duplicate citations were screened. After a thorough review, 980 articles were excluded per title or abstract screen, and 36 articles were selected after reading the abstract for a full-text review. Next, 14 articles were excluded after

full-text screen and data extraction. A risk of bias assessment was performed, and four other studies were excluded. Finally, 14 studies were included in the final analysis. The search is described in [Figure 1](#).

Risk of bias assessment

All studies included in this analysis underwent assessment using the NOS criteria, with the exception of the study by Ruska et al. (2014)¹⁷ due to its randomized design. Among the analyzed studies, eight were classified as having moderate quality, while six were rated as high quality. The specific criteria and total scores for each study are summarized in [Table 1](#).

Patients baseline characteristics

The systematic review analyzed data from 14 studies comprising a total of 1,166 patients who underwent CAS with TRA. The included studies consisted of both retrospective and prospective designs, with a mean age ranging from 53 to 72 years. [Table 1](#) provides an overview of the key patient characteristics observed, including the mean age of the patients, rates of crossover to other access methods, failure of radial or subclavian artery cannulation, eligibility of the radial artery for the procedure, localization of stenosis in the carotid artery, and the most commonly used stent ([Table 2](#)).

Outcomes

[Table 3](#) presents a summary of the main outcomes, including procedural success rate, crossover, complications such as asymptomatic radial artery occlusion (ARAO), forearm hematoma or ecchymosis, and cerebral vascular attacks (CVA) within 30 days, as well as mortality. Crossover to TFA encompasses cases related to cannulation or radial access failure.

Pooled analysis

The procedural success rate in the majority of the studies included in this analysis was high, with 13 out of 14 studies reporting a success rate of 95% (with a 95% confidence interval ranging from 92% to 98%). [Figure 2](#) illustrates the distribution of these success rates, showing a high level of consistency across the studies. The pooled analysis, using a random-effects model, yielded an overall procedural success rate of 95%. However, it is important to note that there was a moderate level of heterogeneity among the studies, with an I^2 value of 77%.

Crossover to TFA access was observed in 12 of the included studies, with a crossover rate of 6% (with a 95% confidence interval ranging from 3% to 9%). [Figure 3](#) illustrates the distribution of these crossover rates, indicating a moderate level of heterogeneity among the studies ($I^2 = 85\%$) when analyzed using a random-effects model.

The primary reasons for crossover to TFA access were radial access failure and cannulation failure during TRA. Radial access failure was reported in four studies, with a pooled analysis indicating a 0% rate (with a 95% confidence interval ranging from 0%–0%). [Figure 4](#) visually represents the absence of radial access failure based on the

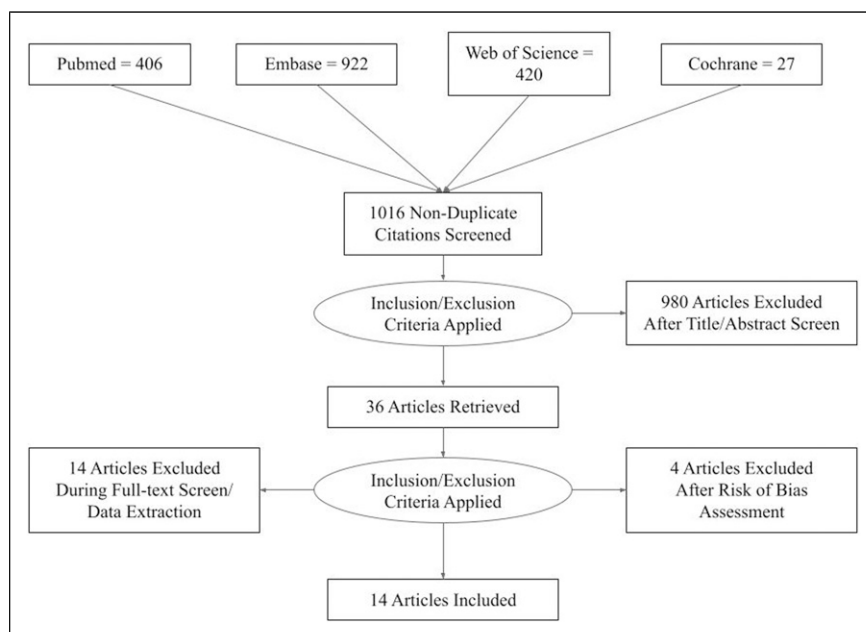


Figure 1. PRISMA flow diagram.

Table 1. Newcastle-Ottawa scale.

Study	Selection				Comparability		Outcome			Total
	Representativeness of exposed cohort	Selection of nonexposed cohort	Ascertainment of exposure	Outcome not presented at the start of the study	Main factor	Additional factor	Assessment of outcomes	Length of follow-up	Adequacy of follow-up	
Iwata 2019 ¹⁸	0	0	1	1	0	0	1	1	1	5
Di Gioia 2023 ¹⁹	1	0	1	1	1	0	1	1	1	7
Etxegoien 2012 ²⁰	1	0	1	1	1	1	1	1	1	8
Erben 2021 ²¹	0	1	1	1	0	0	1	1	1	6
Folmar 2007 ¹⁶	1	0	1	1	0	0	1	1	1	6
Gao 2017 ²²	0	0	1	1	0	0	1	1	1	5
Heck 2022 ²³	1	0	1	1	1	1	1	1	1	8
Hanaoka 2020 ²⁴	1	1	1	1	1	1	1	1	1	9
Kuhn 2020 ²⁵	1	0	1	1	0	0	1	0	0	4
Lee 2016 ¹⁵	1	0	1	1	0	0	1	1	1	6
Mendiz 2016 ²⁶	1	1	1	1	1	0	1	1	1	8
Patel 2010 ²⁷	0	0	1	1	1	0	1	1	1	6
Pinter 2007 ¹⁴	0	0	1	1	0	0	1	1	1	4

pooled analysis, with minimal heterogeneity among the studies ($I^2 = 4\%$).

On the other hand, cannulation failure during TRA was reported in multiple studies, resulting in an overall rate of 4%

(with a 95% confidence interval ranging from 2% to 7%). [Figure 5](#) illustrates the heterogeneity among the studies ($I^2 = 82\%$) when analyzing the rate of cannulation failure using a random-effects model.

Table 2. Main characteristics of included patients.

Study	N	Type of study	Mean age (SD)	Crossover	Radial/SCA failure ^a	Cannulation failure ^b	Radial artery eligibility	Stenosis localization ACI	Stent (most common)
Iwata 2019 ¹⁸	26	Retro	60 - 88	NA	NA	NA	Negative Allen test	Right: 15 Left: 11	Wallstent or Precise stent
Di Gioia 2023 ¹⁹	51	Prosp	67 - 77	Proximal radial (2) Femoral (4)	NA	NA	NA	Right: 34 Left: 17	Wallstent (24), MER (27), and X-act (1)
Etzegoien 2012 ²⁰	382	Retro	68 (0.5)	35	0/35	35/35	NA	Right: 216 Left: 166	Xact (177), Wallstent (91), Precise (45), Xpert (13), Acculink (10), Cristallo Ideale (9), and NexStent (2)
Erben 2021 ²¹	213	Prosp	68.6 (7.9)	0	0	0	NA	Right: 134 Left: 79	NA
Folmar 2007 ¹⁶	42	Prosp	71 (1.26)	7	0/7	7/7	NA	Right: 29 Left: 13	NA
Gao 2017 ²²	28	Retro	72 (5.5)	0	0	0	Positive Allen test	Right: 16 Left: 12	Crital Ideale (Invatec)
Heck 2022 ²³	20	Retro	NA	0	0	0	NA	Right: 14 Left: 6	Carotid Wallstent (Boston Scientific)
Hanaoka 2020 ²⁴	75	Retro	NA	2	0/2	2/2	Negative Allen test	Right: 69 Left: 6	Precise (69), Wallstent (6)
Kuhn 2020 ²⁵	20	Retro	53 - 87	2	1/2	1/2	NA	Right: 22 Left: 0	Wallstent (19), Viabahn stent graft (2), Precise stent (1)
Lee 2016 ¹⁵	38	Retro	69 (10)	NA	NA	NA	Allen test or Doppler ultrasound	NA	Wallstent
Mendiz 2016 ²⁶	101	Retro	71 (10.6)	5	0/5	5/5	Positive Allen test	Right: 57 Left: 44	Acculink (Guidant)
Patel 2010 ²⁷	20	Prosp	65 (5)	4	0/4	4/4	Vascular Doppler	Right: 12 Left: 8	NA
Pinter 2007 ¹⁴	20	Retro	72	2	1/2	1/2	Positive Allen test	Right: 12 Left: 8	Zilver (Cook)
Ruzsa 2014 ¹⁷	130	Prosp	66.8 (8.9)	13	7/13	6/13	Positive Allen test	Right: 73 Left: 57	Wallstent (Boston)

ARAO: asymptomatic radial artery occlusion; NA: Not available; Retro: retrospective; Prosp.: prospective.

^aProcedural failure due to failed radial artery access.

^bProcedural failure due to inability to cannulate the affected common carotid artery with the guide catheter/sheath.

The rate of ARAO was found to be 2% based on data from eight out of the 14 included studies (with a 95% confidence interval ranging from 0% to 5%). Figure 6 illustrates the distribution of these rates, showing a moderate level of heterogeneity among the studies ($I^2 = 67\%$) when analyzed using a random-effects model. Forearm hematoma, a potential complication associated with the TRA, was reported in 10 studies. Among these cases, a pooled analysis revealed a 1% rate (with a 95% confidence interval ranging from 0% to 2%), indicating a low occurrence (Figure 7). Notably, there was minimal heterogeneity among the studies ($I^2 = 0\%$).

The rate of CVA within 30 days was assessed in 13 studies, and the analysis revealed a 2% occurrence (with a 95% confidence interval ranging from 1%–2%). Figure 8 provides a visual representation of the distribution of these rates, indicating a low level of heterogeneity among the studies ($I^2 = 19\%$).

Deaths were reported in three studies, with five deaths observed. However, none of these deaths were directly associated with the TRA.

Discussion

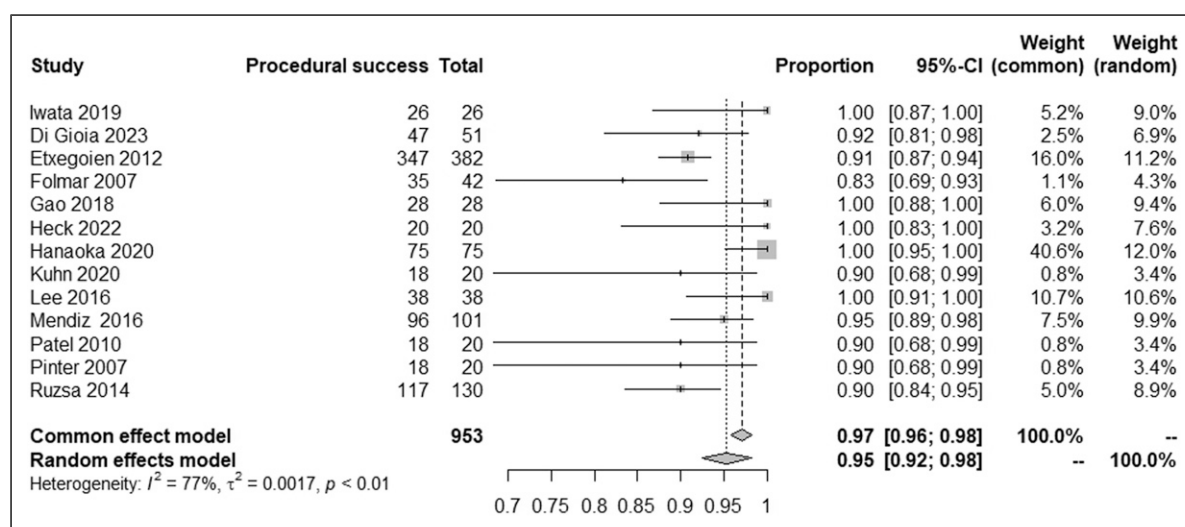
This systematic review with a pooled analysis assessed the safety and efficacy of the TRA approach for carotid stenosis treatment. The analysis revealed a high procedural success rate, decreased risk of bleeding, improved catheter stability, reduced need for deep sedation, and lower complication rates, including ARAO and forearm hematoma. These results support TRA as a safe and effective alternative to the femoral approach, with the potential benefits of fewer complications and improved patient outcomes.

Carotid stent implantation traditionally relies on the femoral artery as the standard access route. However, it presents limitations and challenges.¹⁶ Complications such as access site bleeding and vascular complications are more prevalent, particularly in patients with peripheral artery disease.^{28,29} Additionally, aortic atherosclerotic disease and complex anatomical variations of the aortic and cervical arteries can also pose significant obstacles during the procedure.^{16,30} An earlier comprehensive review analyzed 21 studies, which comprised 1,342 patients in total. This review reported a mere two instances (0.15%) of major

Table 3. Main outcomes.

Study	N	Procedural success	Crossover			Complications			
			CO	CO due to radial access failure	CO due to cannulation failure	ARAO	Forearm hematoma or ecchymosis	CVA in 30 days	Mortality
Iwata 2019 ¹⁸	26	26	NA	NA	NA	0	0	1	0
Di Gioia 2023 ¹⁹	51	47	4	2	2	1	0	0	0
Etzegoien 2012 ²⁰	382	347	35	0	35	23	NA	15	1
Erben 2021 ²¹	213	NA	0	0	0	NA	NA	6	2
Folmar 2007 ¹⁶	42	35	7	0	7	NA	0	1	0
Gao 2017 ²²	28	28	0	0	0	0	0	0	0
Heck 2022 ²³	20	20	0	0	0	0	1	0	0
Hanaoka 2020 ²⁴	75	75	2	0	2	0	NA	0	0
Kuhn 2020 ²⁵	20	18	2	1	1	NA	0	NA	0
Lee 2016 ¹⁵	38	38	NA	NA	NA	NA	NA	1	0
Mendiz 2016 ²⁶	101	96	5	0	5	NA	1	3	0
Patel 2010 ²⁷	20	18	4	0	4	NA	0	1	0
Pinter 2007 ¹⁴	20	18	2	1	1	1	0	0	0
Ruzsa 2014 ¹⁷	130	117	13	7	6	8	2	1	2

ARAO: asymptomatic radial artery occlusion; CO: crossover; CVA: cerebral vascular attacks.

**Figure 2.** Procedural success rate in carotid artery stenting via transradial approach.

complications and 37 occurrences (2.75%) of minor complications. For successful completion of the procedure, 64 patients (4.77%) needed to switch to the transfemoral approach. Access failure led seven patients (0.52%) to make this switch, while difficulties in cannulating the target vessel caused 57 patients (4.24%) to resort to the transfemoral method.¹⁰

In proper scenarios, TRA provides considerable advantages over femoral access, considering the easily accessible site, the lower incidence of vascular complications, the reduced risk of bleeding, and the earlier mobilization. Previous studies, as Snelling et al., which reported that in 147 patients (99.3%) the catheterization of the radial artery was accomplished successfully,⁷ and Zussman et al., which showed that TRA were successfully achieved for cerebral angiography in

44 patients (88%), supports our results of good outcomes of using the TRA.³¹ This approach also offers the dual benefit of decreased reliance on deep sedation, enhancing patient comfort, and mitigating risks associated with anesthesia.^{32,33} Furthermore, previous studies have suggested comparable results for TRA and TFA in terms of safety and efficacy, proving the applicability of TRA as an alternative.^{34,35}

This pooled analysis revealed success rates (95%) comparable to those reported in large-scale transfemoral studies^{22,36,37} and superior to the preceding review (90.8%).³⁸ Regarding complications, we achieved an incidence of 2% for ARAO and 1% for forearm hematoma. These rates align with the findings of the coronary literature and are inferior to the values reported by Jaroengarmsamer (5.9% for ARAO and 1.4% for forearm hematoma).

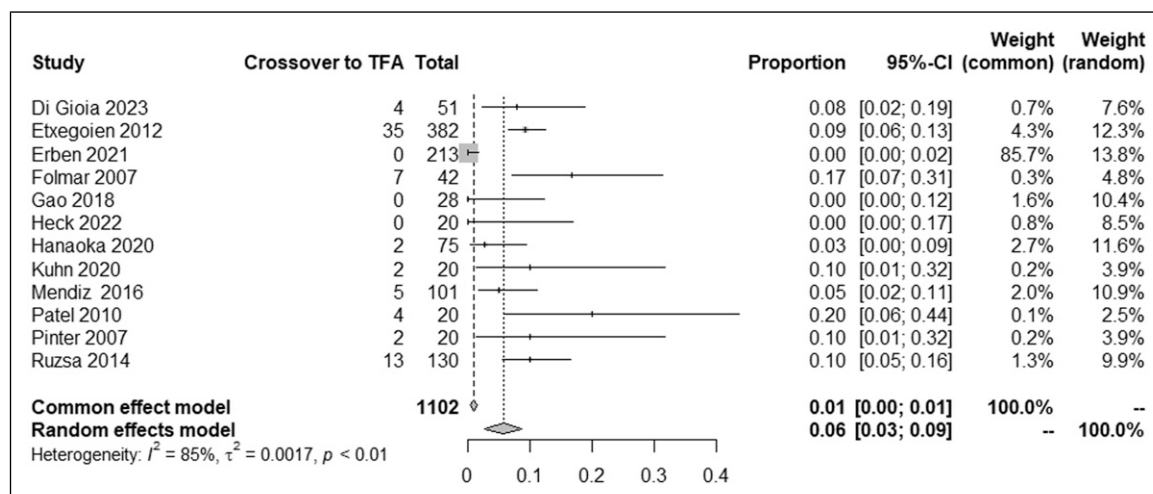


Figure 3. Crossover from transradial to transfemoral access in carotid artery stenting.

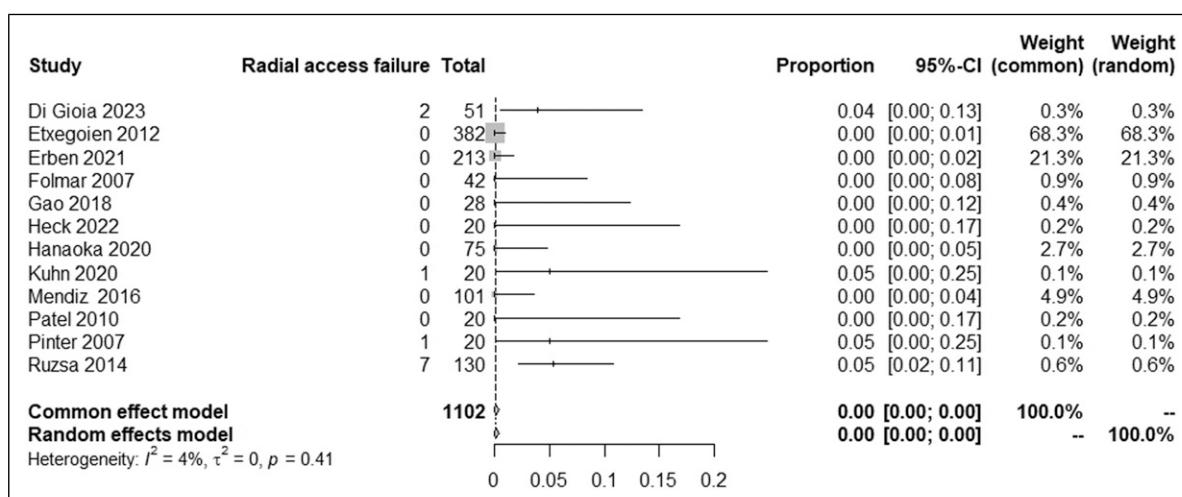


Figure 4. Crossover from transradial to transfemoral access due to radial access failure in carotid artery stenting.

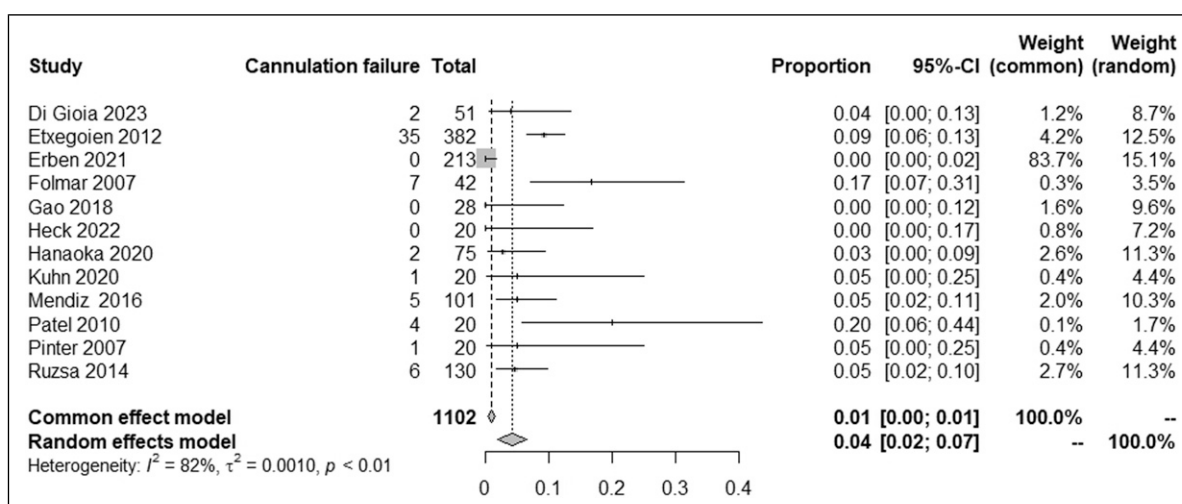


Figure 5. Crossover from transradial to transfemoral access due to cannulation failure in carotid artery stenting.

However, it's important to highlight that only eight out of the 14 included studies address ARAO, and merely 10 report on forearm hematoma. This introduces uncertainty regarding the true incidence of these complications. In a randomized

controlled trial with patients exhibiting easily palpable radial pulses, conducted by interventionists with extensive experience in over 1,000 successful TRA, radial artery occlusion occurred in 13% of cases. Additionally, in a separate study

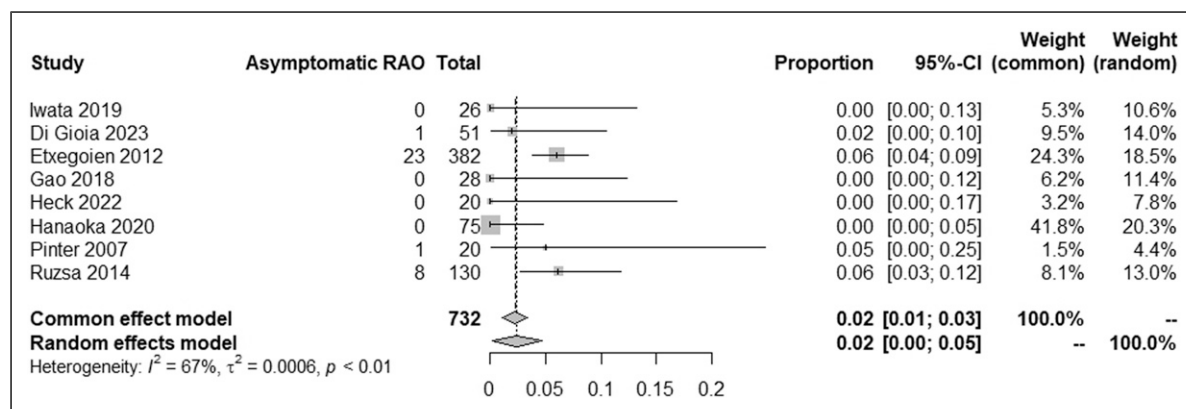


Figure 6. Incidence of asymptomatic radial artery occlusion following transradial approach in carotid artery stenting.

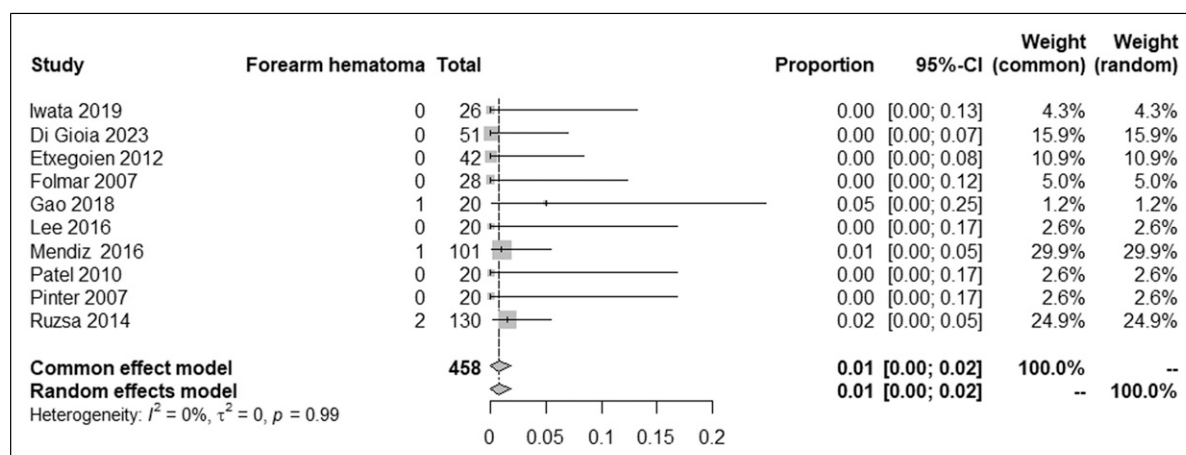


Figure 7. Incidence of forearm hematoma following transradial approach in carotid artery stenting.

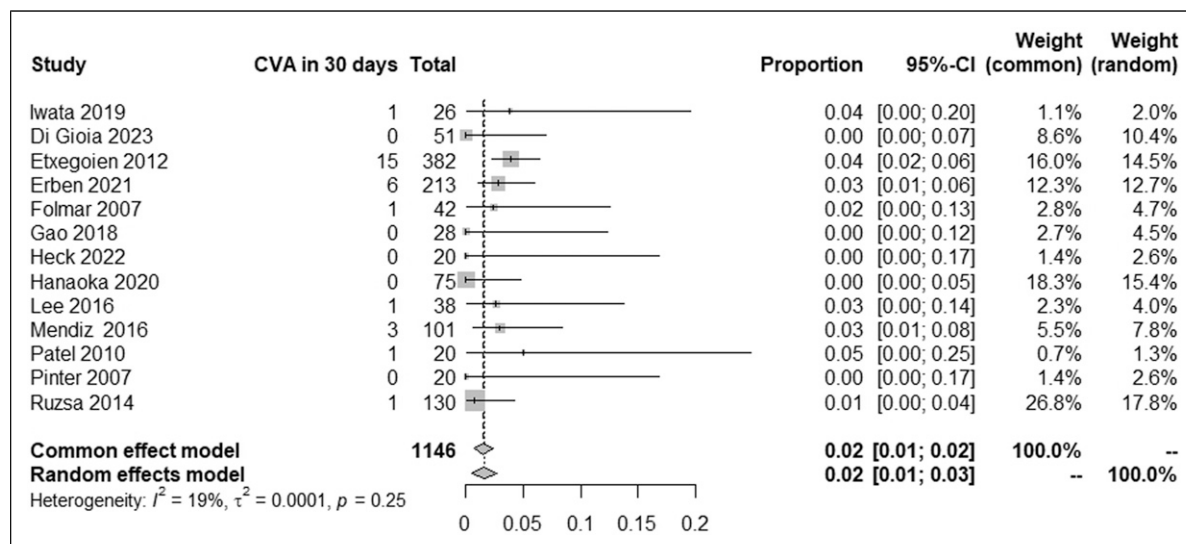


Figure 8. Incidence of cerebral vascular attacks within 30 days following transradial approach in carotid artery stenting.

involving 558 patients who underwent percutaneous coronary intervention, a 10% incidence of radial artery occlusion and a 15% occurrence of radial artery stenosis were reported. Remarkably, these findings not only present a stark contrast to our own results but also offer valuable points of comparison, even though they pertain to cardiac procedures.^{39,40} Although the previous review did not specifically evaluate

CVA within 30 days, our results unveiled a lower occurrence rate (2%) than the 30-day stroke rate (3.5%) in TFA for CAS.⁴¹

Despite the potential benefits, it is crucial to acknowledge the challenges and limitations associated with TRA. Firstly, mastering the technique involves a learning curve, which may require specialized training and experience for optimal

outcomes.^{42–45} Moreover, TRA warrants attention and careful consideration, especially in cases of radiation-induced carotid stenosis, which presents unique anatomical and pathological characteristics compared to typical atherosclerotic lesions.^{46,47} Finally, although the incidence of ARAO and forearm hematomas is relatively low, their clinical significance should be carefully considered in relation to the overall benefits of TRA.

It is important to recognize that TRA may not be the appropriate choice for all patients. Specific contraindications exist that could limit the applicability of TRA. For instance, patients who have a small radial artery-to-sheath size ratio, TRA might prove challenging. Rashid et al. regarding percutaneous cardiac interventions and cardiac angiograms demonstrated that using a smaller introducer sheath is predictive of lower rates of radial artery occlusion, while the use of a larger diameter sheath and prolonged compression duration was associated with higher occlusion rates.⁴⁸ Similarly, certain anatomical conditions such as Type III aortic arches or bovine aortic arches could impede successful radial access. Additionally, the nuances in approach between right and left-sided carotid interventions further add to the complexity. Therefore, thorough patient assessment, considering these factors, the tailored decision is crucial for determining the best approach for CAS procedures.^{49,50}

Limitations

Our study possesses various limitations. Firstly, incorporating observational studies introduces bias into our analysis, with only one eligible randomized trial. Although we conducted quality assessments to mitigate bias and evaluate the methodological rigor of our studies, limitations associated with the study design persist.

Furthermore, within the pool of 13 observational studies, nine had a retrospective design, further underscoring the potential for publication bias, as it stems from the prevailing retrospective nature of the studies. In addition, specific results exhibited heterogeneity, further complicating the interpretation. Other factors to consider are the lack of available and consistent eligibility criteria for radial access across the studies, and uncertainty regarding the true incidence of ARAO and hematoma complications, necessitating caution in analyzing our findings.

Conclusion

Our findings suggest that the TRA for CAS may offer promising outcomes, characterized by high success rates and low complication rates. This research holds significant potential to influence clinical practice, as it provides robust evidence that may support the use of TRA for CAS. As the medical community continues to seek safer and more effective treatment methods, our findings contribute valuable insights that may underscore the potential advantages of TRA.

However, it's crucial to acknowledge that challenges may persist, including the learning curve associated with the technique and the potential occurrence of radial artery spasms. This underscores the need for specialized training and experience to overcome these challenges and ensure

optimal patient outcomes. Therefore, proper patient selection and careful pre-procedural evaluation are paramount to mitigate risk and optimize the advantages associated with this approach. Our research highlights the importance of these considerations, emphasizing the need for a comprehensive approach to implementing TRA for CAS in clinical practice.

Future research would potentially focus on conducting more randomized, controlled trials to further validate these findings and minimize potential biases. Moreover, studies should aim to establish consistent eligibility criteria for radial access to facilitate more accurate comparisons and analyses. Additionally, research should explore the long-term outcomes of TRA for CAS, including the incidence complications, to better understand the overall benefits and potential risks of this approach.

Declaration of conflicting interests

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